

| Method                | $E$ | $ppl$ | % Knowledge Acquisition Performance ( $\Delta$ ) |                      |                      | Entity Learning |      |        | % Out of Domain Performance ( $\Delta$ ) |                       |                       |
|-----------------------|-----|-------|--------------------------------------------------|----------------------|----------------------|-----------------|------|--------|------------------------------------------|-----------------------|-----------------------|
|                       |     |       | Avg.                                             | High-Freq.           | Low-Freq.            | #Acq            | #Ret | #Forg. | Avg.                                     | MMLU                  | Social-IQA            |
| Base                  | 0   | –     | 52                                               | 60                   | 44                   | –               | –    | –      | 72                                       | 66                    | 78                    |
| LoRA                  | 1   | 7.78  | 52 (= 0)                                         | 62 ( $\uparrow$ 2)   | 42 ( $\downarrow$ 2) | –               | –    | –      | 48 ( $\downarrow$ 24)                    | 50 ( $\downarrow$ 16) | 47 ( $\downarrow$ 31) |
|                       | 2   | 7.16  | 52 (= 0)                                         | 61 ( $\uparrow$ 1)   | 44 (= 0)             | 5               | 35   | 5      | 54 ( $\downarrow$ 18)                    | 47 ( $\downarrow$ 19) | 61 ( $\downarrow$ 17) |
|                       | 3   | 6.81  | 50 ( $\downarrow$ 2)                             | 60 (= 0)             | 40 ( $\downarrow$ 4) | 6               | 32   | 8      | 56 ( $\downarrow$ 16)                    | 53 ( $\downarrow$ 13) | 60 ( $\downarrow$ 18) |
|                       | 4   | 6.54  | 48 ( $\downarrow$ 4)                             | 58 ( $\downarrow$ 2) | 39 ( $\downarrow$ 5) | 1               | 28   | 10     | 53 ( $\downarrow$ 19)                    | 50 ( $\downarrow$ 16) | 56 ( $\downarrow$ 22) |
|                       | 5   | 6.31  | 51 ( $\downarrow$ 1)                             | 60 (= 0)             | 43 ( $\downarrow$ 1) | 11              | 26   | 3      | 52 ( $\downarrow$ 20)                    | 49 ( $\downarrow$ 17) | 55 ( $\downarrow$ 23) |
|                       | 6   | 6.12  | 51 ( $\downarrow$ 1)                             | 60 (= 0)             | 42 ( $\downarrow$ 2) | 6               | 29   | 8      | 50 ( $\downarrow$ 22)                    | 46 ( $\downarrow$ 20) | 54 ( $\downarrow$ 24) |
|                       | 7   | 5.95  | 51 ( $\downarrow$ 1)                             | 61 ( $\uparrow$ 1)   | 42 ( $\downarrow$ 2) | 5               | 30   | 5      | 47 ( $\downarrow$ 25)                    | 43 ( $\downarrow$ 23) | 50 ( $\downarrow$ 28) |
|                       | 8   | 5.80  | 51 ( $\downarrow$ 1)                             | 59 ( $\downarrow$ 1) | 42 ( $\downarrow$ 2) | 4               | 28   | 7      | 46 ( $\downarrow$ 26)                    | 46 ( $\downarrow$ 20) | 46 ( $\downarrow$ 32) |
|                       | 9   | 5.61  | 52 (= 0)                                         | 61 ( $\uparrow$ 1)   | 42 ( $\downarrow$ 2) | 7               | 29   | 3      | 50 ( $\downarrow$ 22)                    | 44 ( $\downarrow$ 22) | 56 ( $\downarrow$ 22) |
|                       | 10  | 5.56  | 51 ( $\downarrow$ 1)                             | 60 (= 0)             | 42 ( $\downarrow$ 2) | 6               | 27   | 9      | 44 ( $\downarrow$ 28)                    | 38 ( $\downarrow$ 28) | 51 ( $\downarrow$ 27) |
| Step-wise Regular.    | 1   | 7.86  | 52 (= 0)                                         | 60 (= 0)             | 43 ( $\downarrow$ 1) | –               | –    | –      | 65 ( $\downarrow$ 7)                     | 59 ( $\downarrow$ 7)  | 70 ( $\downarrow$ 8)  |
|                       | 2   | 7.25  | 52 (= 0)                                         | 62 ( $\uparrow$ 2)   | 43 ( $\downarrow$ 1) | 1               | 34   | 5      | 59 ( $\downarrow$ 13)                    | 52 ( $\downarrow$ 14) | 67 ( $\downarrow$ 11) |
|                       | 3   | 6.91  | 54 ( $\uparrow$ 2)                               | 64 ( $\uparrow$ 4)   | 45 ( $\uparrow$ 1)   | 7               | 33   | 2      | 60 ( $\downarrow$ 12)                    | 55 ( $\downarrow$ 11) | 65 ( $\downarrow$ 13) |
|                       | 4   | 6.65  | 53 ( $\uparrow$ 1)                               | 61 ( $\uparrow$ 1)   | 45 ( $\uparrow$ 1)   | 5               | 35   | 5      | 56 ( $\downarrow$ 16)                    | 56 ( $\downarrow$ 10) | 57 ( $\downarrow$ 21) |
|                       | 5   | 6.44  | 53 ( $\uparrow$ 1)                               | 61 ( $\uparrow$ 1)   | 45 ( $\uparrow$ 1)   | 3               | 36   | 4      | 58 ( $\downarrow$ 14)                    | 54 ( $\downarrow$ 12) | 61 ( $\downarrow$ 17) |
|                       | 6   | 6.27  | 52 (= 0)                                         | 63 ( $\uparrow$ 3)   | 42 ( $\downarrow$ 2) | 3               | 35   | 4      | 55 ( $\downarrow$ 17)                    | 52 ( $\downarrow$ 14) | 57 ( $\downarrow$ 21) |
|                       | 7   | 6.05  | 53 ( $\uparrow$ 1)                               | 61 ( $\uparrow$ 1)   | 45 ( $\uparrow$ 1)   | 5               | 30   | 8      | 54 ( $\downarrow$ 18)                    | 51 ( $\downarrow$ 15) | 58 ( $\downarrow$ 20) |
|                       | 8   | 5.99  | 52 (= 0)                                         | 60 (= 0)             | 44 (= 0)             | 12              | 32   | 3      | 51 ( $\downarrow$ 21)                    | 48 ( $\downarrow$ 18) | 54 ( $\downarrow$ 24) |
|                       | 9   | 5.85  | 55 ( $\uparrow$ 3)                               | 63 ( $\uparrow$ 3)   | 47 ( $\uparrow$ 3)   | 7               | 36   | 8      | 49 ( $\downarrow$ 23)                    | 46 ( $\downarrow$ 20) | 52 ( $\downarrow$ 26) |
|                       | 10  | 5.74  | 54 ( $\uparrow$ 2)                               | 62 ( $\uparrow$ 2)   | 46 ( $\uparrow$ 2)   | 8               | 36   | 7      | 50 ( $\downarrow$ 22)                    | 50 ( $\downarrow$ 16) | 50 ( $\downarrow$ 28) |
| Pre-training Regular. | 1   | 8.19  | 52 (= 0)                                         | 60 (= 0)             | 43 ( $\downarrow$ 1) | –               | –    | –      | 68 ( $\downarrow$ 4)                     | 63 ( $\downarrow$ 3)  | 74 ( $\downarrow$ 4)  |
|                       | 2   | 8.05  | 52 (= 0)                                         | 61 ( $\uparrow$ 1)   | 43 ( $\downarrow$ 1) | 6               | 31   | 4      | 69 ( $\downarrow$ 3)                     | 63 ( $\downarrow$ 3)  | 74 ( $\downarrow$ 4)  |
|                       | 3   | 7.98  | 53 ( $\uparrow$ 1)                               | 61 ( $\uparrow$ 1)   | 45 ( $\uparrow$ 1)   | 5               | 33   | 4      | 69 ( $\downarrow$ 3)                     | 63 ( $\downarrow$ 3)  | 75 ( $\downarrow$ 3)  |
|                       | 4   | 7.95  | 52 (= 0)                                         | 60 (= 0)             | 44 (= 0)             | 2               | 33   | 5      | 71 ( $\downarrow$ 1)                     | 64 ( $\downarrow$ 2)  | 77 ( $\downarrow$ 1)  |
|                       | 5   | 7.92  | 53 ( $\uparrow$ 1)                               | 61 ( $\uparrow$ 1)   | 45 ( $\uparrow$ 1)   | 5               | 34   | 1      | 69 ( $\downarrow$ 3)                     | 64 ( $\downarrow$ 2)  | 75 ( $\downarrow$ 3)  |
|                       | 6   | 7.90  | 53 ( $\uparrow$ 1)                               | 62 ( $\uparrow$ 2)   | 44 (= 0)             | 2               | 35   | 4      | 69 ( $\downarrow$ 3)                     | 64 ( $\downarrow$ 2)  | 74 ( $\downarrow$ 4)  |
|                       | 7   | 7.90  | 52 (= 0)                                         | 61 ( $\uparrow$ 1)   | 43 ( $\downarrow$ 1) | 3               | 34   | 3      | 70 ( $\downarrow$ 2)                     | 65 ( $\downarrow$ 1)  | 75 ( $\downarrow$ 3)  |
|                       | 8   | 7.87  | 53 ( $\uparrow$ 1)                               | 61 ( $\uparrow$ 1)   | 44 (= 0)             | 4               | 33   | 4      | 68 ( $\downarrow$ 4)                     | 64 ( $\downarrow$ 2)  | 72 ( $\downarrow$ 6)  |
|                       | 9   | 7.86  | 53 ( $\uparrow$ 1)                               | 62 ( $\uparrow$ 2)   | 44 (= 0)             | 3               | 33   | 4      | 69 ( $\downarrow$ 3)                     | 65 ( $\downarrow$ 1)  | 73 ( $\downarrow$ 5)  |
|                       | 10  | 7.85  | 53 ( $\uparrow$ 1)                               | 61 ( $\uparrow$ 1)   | 45 ( $\uparrow$ 1)   | 5               | 34   | 2      | 69 ( $\downarrow$ 3)                     | 64 ( $\downarrow$ 2)  | 75 ( $\downarrow$ 3)  |
| Structured Annotation | 1   | 4.68  | 53 ( $\uparrow$ 1)                               | 61 ( $\uparrow$ 1)   | 45 ( $\uparrow$ 1)   | –               | –    | –      | 59 ( $\downarrow$ 13)                    | 55 ( $\downarrow$ 11) | 63 ( $\downarrow$ 15) |
|                       | 2   | 4.38  | 52 (= 0)                                         | 61 ( $\uparrow$ 1)   | 44 (= 0)             | 6               | 32   | 5      | 58 ( $\downarrow$ 14)                    | 54 ( $\downarrow$ 12) | 62 ( $\downarrow$ 16) |
|                       | 3   | 4.22  | 51 ( $\downarrow$ 1)                             | 59 ( $\downarrow$ 1) | 42 ( $\downarrow$ 2) | 4               | 33   | 5      | 57 ( $\downarrow$ 15)                    | 53 ( $\downarrow$ 13) | 61 ( $\downarrow$ 17) |
|                       | 4   | 4.09  | 50 ( $\downarrow$ 2)                             | 60 (= 0)             | 41 ( $\downarrow$ 3) | 2               | 32   | 5      | 51 ( $\downarrow$ 21)                    | 54 ( $\downarrow$ 12) | 48 ( $\downarrow$ 30) |
|                       | 5   | 3.98  | 50 ( $\downarrow$ 2)                             | 57 ( $\downarrow$ 3) | 42 ( $\downarrow$ 2) | 4               | 30   | 4      | 50 ( $\downarrow$ 22)                    | 51 ( $\downarrow$ 15) | 49 ( $\downarrow$ 29) |
|                       | 6   | 3.89  | 47 ( $\downarrow$ 5)                             | 57 ( $\downarrow$ 3) | 38 ( $\downarrow$ 6) | 3               | 29   | 5      | 54 ( $\downarrow$ 18)                    | 50 ( $\downarrow$ 16) | 57 ( $\downarrow$ 21) |
|                       | 7   | 3.79  | 45 ( $\downarrow$ 7)                             | 54 ( $\downarrow$ 6) | 36 ( $\downarrow$ 8) | 3               | 21   | 11     | 49 ( $\downarrow$ 23)                    | 49 ( $\downarrow$ 17) | 48 ( $\downarrow$ 30) |
|                       | 8   | 3.75  | 49 ( $\downarrow$ 3)                             | 58 ( $\downarrow$ 2) | 41 ( $\downarrow$ 3) | 9               | 20   | 4      | 45 ( $\downarrow$ 27)                    | 43 ( $\downarrow$ 23) | 47 ( $\downarrow$ 31) |
|                       | 9   | 3.68  | 49 ( $\downarrow$ 3)                             | 58 ( $\downarrow$ 2) | 41 ( $\downarrow$ 3) | 8               | 23   | 6      | 49 ( $\downarrow$ 23)                    | 44 ( $\downarrow$ 22) | 55 ( $\downarrow$ 23) |
|                       | 10  | 3.62  | 45 ( $\downarrow$ 7)                             | 54 ( $\downarrow$ 6) | 37 ( $\downarrow$ 7) | 6               | 19   | 12     | 43 ( $\downarrow$ 29)                    | 38 ( $\downarrow$ 28) | 48 ( $\downarrow$ 30) |
| Curriculum Learning   | 1   | 7.80  | 51 ( $\downarrow$ 1)                             | 60 (= 0)             | 42 ( $\downarrow$ 2) | –               | –    | –      | 59 ( $\downarrow$ 13)                    | 56 ( $\downarrow$ 10) | 63 ( $\downarrow$ 15) |
|                       | 2   | 7.22  | 53 ( $\uparrow$ 1)                               | 62 ( $\uparrow$ 2)   | 44 (= 0)             | 8               | 31   | 2      | 50 ( $\downarrow$ 22)                    | 54 ( $\downarrow$ 12) | 46 ( $\downarrow$ 32) |
|                       | 3   | 6.74  | 49 ( $\downarrow$ 3)                             | 59 ( $\downarrow$ 1) | 38 ( $\downarrow$ 6) | 2               | 33   | 6      | 58 ( $\downarrow$ 14)                    | 56 ( $\downarrow$ 10) | 60 ( $\downarrow$ 18) |
|                       | 4   | 6.36  | 47 ( $\downarrow$ 5)                             | 57 ( $\downarrow$ 3) | 38 ( $\downarrow$ 6) | 1               | 29   | 6      | 54 ( $\downarrow$ 18)                    | 52 ( $\downarrow$ 14) | 55 ( $\downarrow$ 23) |
|                       | 5   | 6.17  | 49 ( $\downarrow$ 3)                             | 59 ( $\downarrow$ 1) | 39 ( $\downarrow$ 5) | 6               | 27   | 3      | 56 ( $\downarrow$ 16)                    | 52 ( $\downarrow$ 14) | 61 ( $\downarrow$ 17) |
|                       | 6   | 5.93  | 49 ( $\downarrow$ 3)                             | 58 ( $\downarrow$ 2) | 39 ( $\downarrow$ 5) | 4               | 30   | 3      | 45 ( $\downarrow$ 27)                    | 44 ( $\downarrow$ 22) | 46 ( $\downarrow$ 32) |
|                       | 7   | 5.70  | 47 ( $\downarrow$ 5)                             | 56 ( $\downarrow$ 4) | 37 ( $\downarrow$ 7) | 5               | 26   | 8      | 51 ( $\downarrow$ 21)                    | 47 ( $\downarrow$ 19) | 54 ( $\downarrow$ 24) |
|                       | 8   | 5.66  | 48 ( $\downarrow$ 4)                             | 56 ( $\downarrow$ 4) | 39 ( $\downarrow$ 5) | 5               | 25   | 6      | 48 ( $\downarrow$ 24)                    | 45 ( $\downarrow$ 21) | 50 ( $\downarrow$ 28) |
|                       | 9   | 5.42  | 47 ( $\downarrow$ 5)                             | 57 ( $\downarrow$ 3) | 38 ( $\downarrow$ 6) | 5               | 24   | 6      | 49 ( $\downarrow$ 23)                    | 45 ( $\downarrow$ 21) | 52 ( $\downarrow$ 26) |
|                       | 10  | 5.38  | 47 ( $\downarrow$ 5)                             | 56 ( $\downarrow$ 4) | 38 ( $\downarrow$ 6) | 8               | 22   | 7      | 57 ( $\downarrow$ 15)                    | 51 ( $\downarrow$ 15) | 63 ( $\downarrow$ 15) |

Table 5: Early-epoch learning dynamics for LLaMA<sub>8B</sub> across different CPT strategies. We report average entity performance (overall, high-frequency, low-frequency), entities acquired, retained, forgotten, and out-of-domain changes in Avg-OOD, MMLU, and Social-IQA.